Yong Sheng Koay

PhD Candidate, AI For Theoretical High Energy Physics, Uppsala University

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Educational Background

PhD in Physics, *Uppsala University, Sweden — Expected Completion Date : 28 April 2027* **2023 – Present**
Topic: Beyond Standard Model Physics using Machine Learning Techniques.
Supervisors: Stefano Moretti, Rikard Enberg, Jose Eliel Camargo-Molina.

MSc in Physics, *University of Bonn, Germany* **2018 – 2020**
Grade: 1.9/5.0 (gut). Thesis: Constraining the B-anomalies Motivated Leptoquark Model at the LHC.
Supervisor: Manuel Drees.

BSc in Physics, *University of Malaya, Malaysia* **2014 – 2018**
CGPA 3.79/4.00 (Pass with Honours, with Distinction). Thesis: Phase Transitions of Ising Models in Ferromagnets.
Supervisor: Chew Khian Hooi.

Research Experiences

Visiting Researcher, *DAMTP, University of Cambridge, UK* **Oct – Dec 2025**
Steering in Theory Space. Supervisor: Sven Krippendorf.

Student Research Fellow, *Aveiro University, Portugal* **Jun – Oct 2022**
Flavour anomalies in leptoquark and scalar singlet models. Supervisor: Antonio Morais.

Research Assistant, *BCTP, Universität Bonn, Germany* **Jan 2021 – Mar 2022**
Lepton-initiated processes at the LHC; R-parity violating MSSM at the LHC. Supervisor: Manuel Drees.

Research Intern, *BCTP, Universität Bonn, Germany* **Aug – Sep 2019**
Constraining MSSM at the LHC. Supervisor: Manuel Drees.

Research Intern, *Academia Sinica, Taiwan* **Jul – Aug 2018**
Random structure search of high-pressure iron oxides. Supervisors: Hsing Cheng-Rong and Wei Ching-Ming.

Research Assistant, *CTP, University of Malaya, Malaysia* **Feb – Jul 2018**
Ab-initio calculation of graphene quantum dots. Supervisors: Goh Boon Tong and Chew Khian Hooi.

Honours and Awards

Best Poster and Featured Plenary Highlight Talk (top 5 of 162), EuCAIFCon	**2026**
Letterstedt Travel Scholarship (73,649 SEK), Letterstedts Resestipendium	**2025**
Lundström-Åman Scholarship (16,741 SEK), Lundström-Åmans Stipendiestiftelse	**2025**
Anna Maria Lundins Scholarship (29,645 SEK), Småland's Nation, Uppsala	**2024**
DAAD Partial Scholarship, German Federal Foreign Office (AA)	**2020**
Deutschlandstipendium, Bonner Universitätsstiftung	**2018/19**
Public Service Department Scholarship, Malaysia	**2014–18**
2nd Top Presenter, Physics Colloquium, University of Malaya	**2017**
Dean's List Award (3 years), Faculty of Science, University of Malaya	**2014–17**

Publications

Full publication list provided separately. For updated list: https://inspirehep.net/authors/2668337.
5 publications (3 peer-reviewed, 2 submitted), 2 in preparation. INSPIRE (Google Scholar) - citations : 48 (47) | h-index : 3 (2)

Selected Presentations / Seminars

15+ presentations (including 3 invited talks, 1 highlight plenary talk) at international conferences, workshops and seminars. Complete and updated list available at : https://kys-sheng.github.io/talks.html.

10 selected presentations are shown below:

Talk — Steering in Theory Space: Representation Engineering of a Lagrangian Transformer **Sep 2026**
ML4JETS 2026, Vienna, Austria

Talk — Fine-Tunable Region-Conditioned Simulation-Based Inference for Collider Analysis **Sep 2026**
ML4JETS 2026, Vienna, Austria

Highlight Plenary Talk — Steering in Theory Space: Representation Engineering of a Lagrangian Transformer **Aug 2026**
EuCAIFCon 2026, Heidelberg, Germany — Selected plenary highlight talk (top 5 of 162).

Talk — BART-L: Towards a Foundation Model for Theoretical High-Energy Physics **Dec 2025**
String Data 2025, London, UK

Invited Talk — BART-L: Towards a Foundation Model for Theoretical High-Energy Physics **Oct 2025**
DAMTP Data Intensive Science Seminar, University of Cambridge, UK

Invited Workshop (Talk) — AI for Equations **Sep 2025**
Build Big or Build Smart Workshop, MIAPbP, Garching, Germany

Invited Workshop (Talk/Tutorial) — How to do NLP-like Research in Physics **Apr 2025**
AI4Physics Learning Workshop, Uppsala University, Sweden — With E. Camargo-Molina.

Talk — When BART Writes Lagrangians: Towards a HEP-LLM (Large Lagrangian Model) **Oct 2024**
Partikeldagarna 2024, Uppsala, Sweden

Poster — Transforming Particle Theory: Lagrangian Generation through Transformers **Jul 2024**
ICHEP 2024, Prague, Czech Republic

Talk — Searches for the S1 Leptoquark **Mar 2022**
IRN Terascale @ Bonn, Bethe Center for Theoretical Physics (BCTP), Germany

Teaching Experiences

Co-supervision (Planned), *Uppsala University, Sweden* **Nov 2026 – June 2027**
30–45 Credits MSc project: ML for Beyond-Standard Model Physics, with Stefano Moretti.

Teaching Assistant, *Uppsala University, Sweden* **Jan 2024 – Present**
Quantum Field Theory (MSc); Applied Deep Learning in Physics and Engineering (MSc/PhD); Advanced Applied Deep Learning in Physics and Engineering (MSc/PhD).

Teaching Assistant, *Universität Bonn, Germany* **Apr 2020 – Mar 2022**
Advanced Quantum Theory; Advanced Theoretical Particle Physics; Theoretical Particle Physics; General Relativity and Cosmology (all MSc).

Other Experiences

Participant, *MIAPbP "Build Big or Build Smart" Program, Garching, Germany* **Aug – Sep 2025**

Participant + Volunteer, *HIGGS 2024, Uppsala University, Sweden* **Nov 2024**

Participant, *IAIFI PhD Summer School 2024, MIT, USA* **Aug 2024**

Participant, *MITP Summer School 2023: Machine Learning in Particle Theory, Mainz, Germany* **Jul 2023**

Organizing Committee, *BSM^2 BrainStorming Meeting, GR@V group, University of Aveiro, Portugal* **Oct 2022**

Research Exchange Student, *Sakura Exchange Program in Science, Maezono Lab, JAIST, Japan* **Jul 2017**

Computer Skills

Full links and project descriptions at: `https://kys-sheng.github.io/tools.html`

ML Projects	BART-Lagrangian [Model, Dataset, Demo]; Theory Space Navigation [Demo]; RACNN Bump Hunt [Demo]; Interactive Visualization for GNN & Transformers [Teaching Tools];
Non-ML Projects	abc-rpv [Python Library]; MCTS-based Hex game engine
Language/Tools	Python, C++, Mathematica, LaTeX, FORTRAN, PyTorch, Keras, sklearn, anthropic, google-genai, GitHub, HuggingFace, emcee, SLURM, HTCondor, mpi4py || **HEP:** FeynRules, MadGraph, Pythia, Delphes, CheckMATE, SARAH, SPheno, flavio, smelli, wilson, DsixTools, AutoEFT, MadAnalysis, ROOT, LHAPDF

Languages

Native/Bilingual	English (IELTS : Band 8), Mandarin, Hokkien, Malay
Competent / Novice	Cantonese; German (A2), Swedish (A1)

References

Stefano Moretti (Main PhD supervisor, THEP)
Institutionen för fysik och astronomi, Uppsala universitet
Email: stefano.moretti@physics.uu.se

Rikard Enberg (PhD co-supervisor, THEP)
Institutionen för fysik och astronomi, Uppsala universitet
Email: rikard.enberg@physics.uu.se

Eliel Camargo-Molina (PhD co-supervisor, THEP)
Institutionen för speldesign, Uppsala universitet
Email: eliel.camargo-molina@speldesign.uu.se

Sven Krippendorf (Collaborator in ML↔Physics)
DAMTP, University of Cambridge
Email: slk38@cam.ac.uk

Prashant Singh (Collaborator in ML)
Department of Information Technology, Uppsala universitet
Email: prashant.singh@scilifelab.uu.se